

Notes: Belo, Gala, and Li (JFE, 2013)

Government Spending, Political Cycles, and the Cross Section of Stock Returns

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1 Big picture

- **Question.** Do presidential partisan cycles affect the cross section of U.S. stock returns through the government spending channel?
- **Setting.** The paper studies publicly traded U.S. firms from July 1955 to December 2009. The key source of heterogeneity is an industry's exposure to government spending, measured from Bureau of Economic Analysis input-output accounts.
- **Economic motivation.** Government spending is large, about one-fifth of U.S. GDP, and political parties differ in fiscal priorities. If Democrats and Republicans allocate spending differently, firms that sell more directly or indirectly to the government should have different cash-flow prospects across presidencies.
- **Core empirical challenge.** Aggregate government spending is noisy and highly correlated with the business cycle. Democratic presidencies also tend to coincide with stronger GDP growth. A simple aggregate time-series regression could therefore confuse spending effects, macro conditions, and political-cycle effects.
- **Main design.** The authors compare firms with high and low *ex ante* exposure to government spending within the same months and then ask whether their relative stock returns reverse across Democratic and Republican presidencies.
- **Key measurement contribution.** The paper constructs a novel industry-level government exposure measure using input-output tables. The measure includes both direct sales to the government and indirect exposure through upstream production chains.
- **Main baseline result.** During Democratic presidencies, the high-government-exposure portfolio outperforms the low-government-exposure portfolio by about **6.1% per year**. During Republican presidencies, the sign reverses: high-exposure firms underperform low-exposure firms by about **4.8% per year**.
- **Presidential stock-market puzzle connection.** The well-known fact that aggregate stock returns are higher under Democrats is concentrated in government-exposed firms. The Democratic-minus-Republican return difference rises monotonically from **2.6%** for low-exposure industries to **13.5%** for high-exposure industries.
- **Key geography result.** The reversal is stronger among firms headquartered in states receiving high per capita federal spending. In high-federal-spending states, the high-minus-low government exposure spread is **+7.0%** under Democrats and **-7.7%** under Republicans.
- **Controls and robustness.** The pattern survives firm characteristics, political contributions, business-cycle variables, and standard risk-factor models. It is also similar in an extended sample beginning in 1929, although that pre-1955 extension is not fully tradable because the exposure data were not yet public.
- **Abnormal-return result.** A political-cycle trading strategy that goes long high-exposure firms and short low-exposure firms under Democrats, and reverses the position under Republicans, earns about **5.3%** per year. Conditional factor-model alphas are as large as **6.9%** per year.

- **Timing result.** The strategy's abnormal returns are concentrated in years 2 and 3 of presidential terms, not immediately after elections. This timing is more consistent with gradual market underreaction than with immediate repricing of political risk.
- **Cash-flow result.** Government exposure predicts future profitability in the same direction as stock returns: high-exposure firms have higher future ROE under Democrats and lower future ROE under Republicans.
- **Main interpretation.** The evidence supports an expected-cash-flow channel: partisan differences in government spending policy affect the profitability and returns of government-exposed firms. The market appears to underreact to this predictable variation.
- **Main caveat.** The paper is not a randomized experiment. The authors explicitly note that unobserved characteristics correlated with government exposure could still matter. The evidence is therefore strongly suggestive, but the causal interpretation is not airtight.

2 Data and measurement (institutional context and key objects)

2.1 Why government spending exposure is the right object

- The paper does not ask whether the whole stock market rises or falls under one party.
- It asks whether firms that are more economically connected to government demand perform differently when the party controlling the presidency changes.
- This cross-sectional approach is powerful because it compares firms within the same aggregate environment.
- A high-government-exposure industry should benefit more from expansions or reallocations of government demand than a low-government-exposure industry.

Interpretation. The design turns a political macro question into an asset-pricing cross-section question: which firms are more exposed to government demand, and do their returns move predictably across partisan cycles?

2.2 Input-output data and direct plus indirect exposure

- The key data source is the Benchmark Input-Output Accounts from the Bureau of Economic Analysis.
- These accounts report flows of goods and services across industries and to final users such as the government sector, private consumption, and private investment.
- The authors measure government demand at the industry level and map it to publicly traded firms.

- The measure includes federal, state, and local government purchases.

Let A be the input-output coefficient matrix, c^G be the vector of government final demand, and y be the vector of total industry output. The Leontief relation is

$$x^G = (I - A)^{-1}c^G. \quad (1)$$

The industry exposure to government spending is then

$$Gov_i = \frac{x_i^G}{y_i}. \quad (2)$$

Interpretation. x_i^G is the total output from industry i required to satisfy government final demand. It includes both goods sold directly to the government and intermediate inputs sold to other industries that ultimately serve government demand. Gov_i is therefore the fraction of industry output economically tied to government spending.

2.3 Timing and tradability of the exposure measure

- Input-output tables are available for 1947, 1958, 1963, 1967, and then years ending in 2 and 7 after 1967.
- The tables are released to the public with substantial lags.
- The authors update the government exposure measure only when the relevant input-output table becomes publicly available.
- This rule makes the portfolio strategy tradable: investors are not allowed to use future exposure information.
- Because of release lags and accounting-data availability, the main sample begins in July 1955 and ends in December 2009.

Interpretation. The paper is careful about look-ahead bias. The return tests use exposure information that investors could have known at the time of portfolio formation.

2.4 Stock return and accounting data

- Monthly stock returns come from CRSP.
- Firm-level accounting data come from the CRSP/Compustat merged annual industrial files.
- The sample requires monthly returns, SIC code, and market capitalization.
- Financial firms and utilities are excluded because these industries are heavily regulated and may have special links to government policy.
- Industries are mainly identified at the three-digit SIC level, with NAICS matching used for later input-output tables.

Interpretation. The return analysis is a standard CRSP/Compustat asset-pricing sample, but the sorting variable is an external measure of exposure to government demand.

2.5 Descriptive facts about government exposure

- The average industry exposure to government spending is about **13.2%**.
- More than **90%** of industries have less than **30%** of output tied to government demand.
- Some industries are much more exposed, with maximum exposure near **68%** in the summary statistics.
- The exposure distribution is similar across Democratic and Republican presidencies, reducing concern that simple composition changes drive the results.

Interpretation. Most industries are private-demand dominated, but there is enough cross-sectional variation to form economically meaningful high- and low-exposure portfolios.

2.6 Government spending and macro conditions across presidencies

- In the main sample, GDP growth is higher under Democratic presidencies than Republican presidencies: **2.54%** versus **1.50%**.
- In the main sample, government spending growth is also higher under Democrats: **1.53%** versus **0.75%**, though this difference is not statistically strong.
- In the post-WWII sample, government spending growth is much higher under Democrats: **3.53%** versus **0.44%**.
- Government spending volatility is also higher under Democrats in the extended and post-WWII samples.

Interpretation. There is evidence of partisan differences in fiscal policy, but aggregate data alone are not enough because political cycles and business cycles are highly correlated.

2.7 Government exposure portfolios

- Each June, the authors sort stocks into five portfolios using the industry exposure measure available at the end of the prior year.
- Returns are value-weighted and tracked from July of year t to June of year $t + 1$.
- Portfolios are rebalanced annually.
- The low-exposure portfolio has average government exposure of about **1.5%**; the high-exposure portfolio has average exposure of about **29.6%**.

- Portfolio membership is stable. The transition matrix has high diagonal entries, including **94.9%** persistence for the low portfolio and **93.3%** persistence for the high portfolio.

Interpretation. The high-minus-low spread compares firms with substantially different exposure to government demand, while portfolio stability means the strategy is not mechanically driven by excessive turnover.

2.8 Examples of high- and low-exposure industries

- High-exposure industries include guided missiles and space vehicles, shipbuilding and repair, radio and television broadcasting, scientific research and development, and defense-related manufacturing.
- Low-exposure industries include food products, carpets, soft drinks, tobacco products, household appliances, breweries, and entertainment-related industries.

Interpretation. The measure has intuitive economic content. It does not simply label all industrial firms as government-dependent; it identifies industries whose output is directly or indirectly tied to public demand.

3 Models and empirical specifications (all equations plus interpretation)

3.1 Conceptual mechanism

The paper's central economic logic is:

Party in White House $\rightarrow$ government spending policy $\rightarrow$ firm cash flows $\rightarrow$ stock returns. (3)

- Under Democratic presidencies, government spending growth and the allocation of spending are expected to favor firms more exposed to government demand.
- Under Republican presidencies, firms more exposed to government demand may receive weaker demand support or face lower expected profitability relative to low-exposure firms.
- If markets fully anticipate these differences, returns should adjust immediately when political information arrives.
- If markets underreact, abnormal returns may emerge gradually as the spending policies affect realized fundamentals.

3.2 Portfolio return comparison

Let R_{t+1}^H be the high-government-exposure portfolio excess return and R_{t+1}^L be the low-government-exposure portfolio excess return. The basic spread is

$$Spread_{t+1} = R_{t+1}^H - R_{t+1}^L. \quad (4)$$

The main portfolio test compares

$$E[Spread_{t+1} \mid Dem_t = 1] \quad \text{and} \quad E[Spread_{t+1} \mid Rep_t = 1]. \quad (5)$$

Interpretation. A positive spread under Democrats and a negative spread under Republicans is the core signature of a government-spending channel.

3.3 Firm-level Fama–MacBeth return regressions

The authors estimate monthly cross-sectional regressions of returns on government exposure interacted with presidential-party dummies and firm controls. A simplified version is

$$r_{i,t+1} = a_t + b_{D,t}(Gov_{i,t} \times Dem_t) + b_{R,t}(Gov_{i,t} \times Rep_t) + \Gamma'_t X_{i,t} + \varepsilon_{i,t+1}. \quad (6)$$

The reported coefficients are time-series averages of the monthly cross-sectional slopes, with Newey–West corrected p -values.

Controls include:

- firm size;
- book-to-market ratio;
- momentum;
- market beta computed from daily returns within the month;
- corporate political contributions;
- interactions with state-level federal-spending dependence.

Interpretation. This test asks whether government exposure has incremental predictive power beyond standard firm characteristics and direct political connectedness.

3.4 Business-cycle return predictability regressions

To separate political-cycle effects from business-cycle effects, the authors estimate time-series regressions for each government exposure portfolio:

$$R_{p,t+1} = b_{0p}Dem_t + b_{1p}Rep_t + \gamma'_p X_t + u_{p,t+1}, \quad p = 1, \dots, 5. \quad (7)$$

Here X_t includes lagged business-cycle and forecasting variables:

- default spread;
- term spread;
- short-term Treasury bill rate;
- dividend yield;
- inflation;
- unemployment;
- industrial production growth.

Interpretation. If the government exposure spread is only a disguised recession/expansion effect, it should disappear after these controls. It does not.

3.5 Political-cycle investment strategy

The strategy used to study abnormal returns is

$$PC_{t+1} = Dem_t(R_{t+1}^H - R_{t+1}^L) + Rep_t(R_{t+1}^L - R_{t+1}^H). \quad (8)$$

- Under Democrats, the strategy is long high government exposure and short low government exposure.
- Under Republicans, the strategy reverses: long low exposure and short high exposure.
- The average return on this strategy is called the presidential investment premium.

Interpretation. This strategy directly monetizes the return reversal across parties. It turns the conditional pattern into a single return series that can be tested with asset-pricing models.

3.6 Conditional asset-pricing decomposition

The authors follow a conditional-factor model logic. Expected returns are written as

$$E_t R_{t+1} = \alpha + \sum_j \beta_{j,t} \lambda_{j,t}, \quad (9)$$

with time-varying betas

$$\beta_{j,t} = \beta_{j,0} + \beta'_{j,1} Z_t. \quad (10)$$

The empirical regression is

$$R_{t+1} = \alpha + \sum_j \beta_{j,t} F_{j,t+1} + \epsilon_{t+1}. \quad (11)$$

The factor models include:

- the CAPM market factor;
- the Fama–French three factors: market, SMB, and HML;

- the Carhart four factors: market, SMB, HML, and momentum.

The conditioning variables Z_t are considered in three versions:

- no instruments: unconditional factor model;
- business-cycle instruments: dividend-price ratio, default spread, term spread, and risk-free rate;
- business-cycle instruments plus a Democratic-president dummy.

Interpretation. If the presidential investment premium is just compensation for exposure to standard risks, the alpha should be zero. The paper finds that alphas remain positive and often large.

3.7 Within-presidency alpha specification

To test whether returns arrive immediately after election-related political uncertainty is resolved, the alpha is allowed to vary by year within the presidential term:

$$\alpha_t = \sum_{k=1}^4 \alpha_k Year(k)_t. \quad (12)$$

Interpretation. A pure political-risk repricing story might predict large returns early in the term. Instead, the paper finds stronger abnormal returns in years 2 and 3, consistent with gradual learning or underreaction.

3.8 Expected profitability regression

To study the cash-flow channel, the authors estimate annual Fama–MacBeth regressions of future return on equity:

$$\begin{aligned} ROE_{i,t+1} = & a_0 + a_1 \frac{V_{i,t}}{A_{i,t}} + a_2 DD_{i,t} + a_3 \frac{D_{i,t}}{B_{i,t-1}} + a_4 ROE_{i,t} \\ & + a_5 (Gov_{i,t} \times Dem_t) + a_6 (Gov_{i,t} \times Rep_t) + e_{i,t+1}. \end{aligned} \quad (13)$$

Where:

- ROE is net income scaled by book equity;
- V/A is market value of assets over book assets;
- DD is a non-dividend-paying dummy;
- D/B is dividends over book equity;
- lagged ROE captures profitability persistence.

Interpretation. If stock returns are driven by expected cash flows, government exposure should predict future profitability in the same direction as returns.

3.9 Profitability-volatility regression

The cash-flow risk channel is examined using the squared residuals from the profitability regression:

$$e_{i,t+1}^2 = a_0 + a_1(Gov_{i,t} \times Dem_t) + a_2(Gov_{i,t} \times Rep_t) + z_{i,t+1}. \quad (14)$$

Interpretation. A risk-based explanation requires the volatility of high-exposure firms' cash flows to line up with the return pattern. The evidence is only partly consistent with this: volatility rises with government exposure under Democrats, but it also rises under Republicans, where the return pattern goes the other way.

4 Identification and empirical design (what makes the estimates credible)

4.1 What identifies the main estimate

The core identifying variation is cross-sectional and conditional on time:

- different industries have different predetermined exposure to government spending;
- all firms are observed in the same macro months;
- the high-minus-low return spread flips sign depending on whether the president is Democratic or Republican.

Identification logic. If high-exposure firms always had higher returns, the result could simply reflect a static industry premium. If Democrats always had higher market returns, the result could simply reflect the presidential stock-market puzzle. The paper's distinctive evidence is the interaction: government exposure matters in opposite directions across parties.

4.2 Why the exposure measure is useful

- It is based on real input-output links, not estimated return betas.
- It captures direct and indirect dependence on government final demand.
- It is measured before returns are realized, using public release dates.
- It is stable over time, making it less likely that results come from mechanical reclassification.

Economic message. The exposure variable has a clear production-side interpretation. It is not constructed from the same stock-return data that it later explains.

4.3 Business-cycle confounding

A major threat is that Democrats and Republicans preside over different macroeconomic conditions. The paper addresses this in three ways:

- it compares high- and low-exposure firms within the same month;
- it controls for standard financial predictors and macro variables;
- it shows that the presidential return difference grows with government exposure rather than appearing uniformly across all portfolios.

Economic message. The results are not just that the market performs better under Democrats. They are that firms more exposed to government spending account for a disproportionate part of that difference.

4.4 Firm-characteristic confounding

Government-exposed firms might differ from other firms in size, value, momentum, beta, leverage, or political connectedness. The paper addresses this using:

- Fama–MacBeth firm-level regressions;
- standard return predictors such as size, book-to-market, momentum, and beta;
- corporate political contribution controls;
- geographic interactions based on state-level federal spending.

Economic message. The government exposure effect is not simply a repackaged size, value, momentum, or political-contribution effect.

4.5 Geographic validation

The paper asks whether the government-exposure effect is stronger where federal spending is more important.

- Firms are classified by headquarter state.
- States are split into high and low per capita federal-spending states.
- The high-minus-low government exposure return spread is much stronger in high-federal-spending states.

Economic message. This is an important validation test. If government spending is the channel, then both industry exposure and geographic dependence on federal funds should matter.

4.6 Risk versus underreaction

The paper distinguishes risk compensation from abnormal returns using factor models and cash-flow evidence.

- Standard factor models do not explain the political-cycle investment premium.
- Conditional risk specifications reduce but do not eliminate abnormal returns.
- Abnormal returns are concentrated in years 2 and 3, not only immediately after elections.
- Future profitability moves in the same direction as returns.
- Profitability volatility does not fully match the required risk story, especially under Republicans.

Economic message. The combined evidence is more consistent with market underreaction to predictable government-spending effects than with simple exposure to standard risk factors.

4.7 What the paper cannot fully prove

- It cannot randomly assign government spending exposure to firms.
- It cannot observe all details of government spending allocations to public versus private firms.
- It cannot completely rule out an omitted political risk factor.
- Like most asset-pricing tests, the abnormal-return evidence faces the joint-hypothesis problem: an alpha can reflect mispricing or a missing risk model.
- The authors are careful to describe the evidence as an initial and suggestive study of a new return determinant, rather than a definitive causal experiment.

Takeaway. The evidence is credible because many independent patterns line up, but the causal claim remains qualified.

5 Tables and figures

5.1 Table 1: Summary statistics

Read.

- The average number of firms is about **3,406** per month.
- The value-weighted market excess return is **10.6%** per year across all years.
- Market excess returns are much higher under Democrats: **15.3%** versus **7.7%** under Republicans.
- Mean government exposure is about **13.2%** and is nearly identical across Democratic and Republican presidencies.

- The 90th percentile of government exposure is around **27.4%**; the maximum is about **68.1%**.

Economic message. The exposure distribution itself does not mechanically shift across parties. This helps the paper interpret later return differences as conditional effects rather than simple changes in the types of firms available in each presidency.

Table 1

Summary statistics.

This table reports the summary statistics of selected variables across all years, as well as across Democratic and Republican presidencies. It reports selected moments of the measure of industry exposure to government spending including mean and standard deviation (S.D.). "10th", "50th", "90th" and "Max" are the corresponding average percentiles of the cross-sectional distribution of the government exposure measure. "Firms" is the average number of firms in each month, and "Ret" is the annualized value-weighted average excess return. The data are from July 1955 to December 2009.

President	Firms	Ret	Exposure to government spending (percent)					
			Mean	S.D.	10th	50th	90th	Max
All	3406	10.6	13.2	11.5	1.5	11.1	27.4	68.1
Republican	3340	7.7	13.3	11.7	1.4	11.3	28.4	67.6
Democratic	3511	15.3	13.1	11.2	1.5	10.8	25.9	68.8

Table 2

Government spending across presidencies.

This table reports the mean and standard deviation of selected macroeconomic variables across Democratic (Dem) and Republican (Rep) presidencies. ΔGDP is the growth rate of real per capita gross domestic product, and ΔG is the growth rate of per capita total government spending. "Dif Dem-Rep" is the difference between the corresponding statistic under Democratic and Republican presidencies, and "Dif p -value" is the corresponding p -value (in parenthesis). The annual data ranges are from 1955 to 2009, 1929 to 2009, and 1947 to 2009.

Variables	Mean				Standard deviation			
	Dem	Rep	Dif Dem- Rep	Dif <i>p</i> - value	Dem	Rep	Dif Dem- Rep	Dif <i>p</i> - value
Main sample 1955–2009								
ΔGDP	2.54	1.50	1.04	(0.05)	2.06	2.19	−0.13	(0.79)
ΔG	1.53	0.75	0.78	(0.40)	2.26	2.13	0.13	(0.74)
Extended sample 1929–2009								
ΔGDP	3.50	0.58	2.91	(0.04)	5.30	4.06	1.24	(0.10)
ΔG	4.15	0.61	3.54	(0.45)	24.80	3.19	21.61	(0.00)
Post-WWII sample 1947–2009								
ΔGDP	2.62	1.53	1.09	(0.02)	2.33	2.29	0.04	(0.91)
ΔG	3.53	0.44	3.10	(0.10)	6.64	2.89	3.75	(0.00)

5.2 Table 2: Government spending across presidencies

Read.

- In the main 1955–2009 sample, real per capita GDP growth is **2.54%** under Democrats and **1.50%** under Republicans.
- Main-sample government spending growth is **1.53%** under Democrats and **0.75%** under Republicans.
- In the extended 1929–2009 sample, government spending growth is **4.15%** under Democrats and **0.61%** under Republicans, with much higher volatility under Democrats.
- In the post-WWII sample, government spending growth is **3.53%** under Democrats and **0.44%** under Republicans.

Economic message. The macro evidence is consistent with partisan differences in government spending, but it also shows why identification is hard: political cycles are strongly related to macro conditions.

5.3 Table 3: Portfolio transition frequencies

Read.

- The government exposure portfolios are very stable.

- The low portfolio remains low **94.9%** of the time.
- The high portfolio remains high **93.3%** of the time.
- Even the middle portfolios have high persistence, generally near or above **88%** on the diagonal.

Economic message. The main strategy is not a high-turnover trading rule based on unstable classifications. Exposure to government demand is a persistent industry characteristic.

Table 3
Portfolio transition frequencies.

This table reports the annual transition frequencies (in percentage) across the five government exposure portfolios. The data are from July 1955 to December 2009.

Sort(<i>t</i>)	Sort(<i>t</i> + 1)				
	Lo	2	3	4	Hi
Lo	94.9	3.2	1.0	0.6	0.4
2	3.6	88.4	5.2	1.5	1.3
3	0.7	2.8	91.2	3.3	1.9
4	0.6	0.9	3.9	90.1	4.6
Hi	0.3	1.0	1.1	4.4	93.3

Table 4
Industries with high and low exposure to the government sector.
This table reports a sample of industries in the high and low government exposure portfolios. "Gov expo" is the industry-level measure of exposure to government spending (in percent). The data and the industry classification are based on the 2002 Benchmark Input-Output Accounts table.

I-O code	Industry	Gov expo	Portfolio
Industries with high exposure to the government sector			
336414	Guided missile and space vehicle manufacturing	94.7	High
336611	Shipbuilding and repairing	67.3	High
515100	Radio and television broadcasting	54.7	High
541700	Scientific research and development services	47.0	High
335110	Electric lamp bulb and part manufacturing	45.9	High
211000	Oil and gas extraction	39.9	High
511110	Newspaper publishers	28.3	High
334418	Printed circuit assembly manufacturing	24.0	High
334220	Broadcast and wireless communications equipment	22.6	High
322120	Paper mills	20.6	High
Industries with low exposure to the government sector			
311225	Fats and oils refining and blending	2.9	Low
314110	Carpet and rug mills	2.9	Low
311410	Frozen food manufacturing	1.8	Low
311820	Cookie, cracker, and pasta manufacturing	1.7	Low
339910	Jewelry and silverware manufacturing	1.2	Low
312110	Soft drink and ice manufacturing	1.0	Low
335224	Household laundry equipment manufacturing	0.8	Low
312120	Breweries	0.8	Low
312240	Tobacco product manufacturing	0.4	Low
713950	Bowling centers	0.0	Low

5.4 Table 4: Industries with high and low exposure to government spending

Read.

- High-exposure examples include guided missile and space vehicle manufacturing with **94.7%** government exposure and shipbuilding and repairing with **67.3%**.
- Radio and television broadcasting, scientific R&D services, and some communications-equipment industries also appear in the high portfolio.
- Low-exposure examples include soft drinks, breweries, tobacco products, household laundry equipment, and bowling centers.

Economic message. The variable captures plausible government-demand exposure. It separates defense, infrastructure, research, and communications-related industries from private-consumption industries.

5.5 Table 5: Average characteristics of the government exposure portfolios

Read.

- The low portfolio has average government exposure of **1.51%**; the high portfolio has **29.58%**.
- The portfolios contain many firms and industries, not only one narrow sector.
- Firm size is broadly similar across portfolios.

- Low-exposure portfolios tend to have somewhat higher book-to-market ratios and leverage.

Economic message. The high-minus-low return spread compares portfolios with large exposure differences, while differences in common characteristics are modest enough to be controlled directly in regressions.

Table 5

Average characteristics of the government exposure portfolios.

This table reports the time series averages of median characteristics of five government exposure portfolios. "Market share" is the portfolio market share (in percent) across all firms in the Center for Research in Security Prices, "Firms" is the monthly number of firms in each portfolio, "Industries" is the monthly number of three-digit standard industrial classification industries in each portfolio, "Gov" is the portfolio level of industry exposure to government spending (in percent), "Size" is the (log) market capitalization, "BM" is the ratio of book-to-market equity, and "Lev" is the leverage ratio. The sample is from July 1955 to December 2009.

Portfolio	Market share	Firms	Industries	Gov	Size	BM	Lev
Lo	10.60	543	115	1.51	4.00	0.86	0.40
2	9.02	378	81	4.81	3.95	0.85	0.39
3	22.40	750	121	9.17	4.08	0.79	0.37
4	20.69	719	104	14.74	4.17	0.77	0.34
Hi	12.92	836	97	29.58	3.94	0.71	0.30

Table 6

Average returns of the government exposure portfolios.

This table reports the time series average of the excess value-weighted annualized returns of the five government exposure portfolio across all years, as well as across Democratic (Dem) and Republican (Rep) presidential terms. "Dif Dem-Rep" is the difference in the portfolio average excess returns across Democratic and Republican presidential terms, and "Dif p-value" is the corresponding Newey and West corrected p-value (in parenthesis). "Hi-Lo" is the return of the high minus low government exposure spread portfolio. The sample is from July 1955 to December 2009 and from July 1929 to December 2009.

Portfolio	All years	Dem years	Rep years	Dif Dem-Rep	Dif p-value
Sample period: 1955–2009					
Lo	7.13	8.72	6.14	2.58	(0.59)
2	5.49	7.83	4.03	3.80	(0.42)
3	5.94	9.94	3.45	6.49	(0.10)
4	4.88	10.17	1.58	8.59	(0.05)
Hi	6.54	14.86	1.36	13.50	(0.02)
Hi-Lo	−0.59	6.14	−4.78	10.92	(0.01)
p-value	(0.75)	(0.06)	(0.02)		
Extended sample period: 1929–2009					
Lo	7.26	10.71	3.72	6.99	(0.08)
2	7.12	12.20	1.89	10.32	(0.03)
3	6.84	12.30	1.22	11.09	(0.01)
4	6.94	13.57	0.11	13.46	(0.00)
Hi	8.32	17.04	−0.66	17.70	(0.00)
Hi-Lo	1.06	6.33	−4.38	10.71	(0.00)
p-value	(0.54)	(0.01)	(0.07)		

5.6 Table 6: Average returns of the government exposure portfolios

Read.

- In the main sample, the unconditional high-minus-low spread is only **-0.59%** per year and is not significant.
- Under Democratic presidencies, the high-minus-low spread is **+6.14%** per year.
- Under Republican presidencies, the high-minus-low spread is **-4.78%** per year.
- The difference between the Democratic and Republican spreads is **10.92%** per year with a reported p-value of **0.01**.
- The Democratic-minus-Republican return difference increases monotonically across portfolios, from **2.58%** for low exposure to **13.50%** for high exposure.
- In the extended 1929–2009 sample, the high-minus-low spread is **+6.33%** under Democrats and **-4.38%** under Republicans.

Economic message. This is the paper's central result. Government exposure does not earn a constant premium; it earns a party-contingent premium that flips sign across presidential cycles.

5.7 Figure 1: Government exposure spread across presidential terms

Read.

- The figure plots the high-minus-low government exposure spread for each presidential term in the extended sample.
- Most Democratic terms show positive spreads.
- Most Republican terms show negative spreads.
- There are exceptions, which is natural because stock returns are noisy and individual administrations differ.

Economic message. The result is not driven by only one presidency. The pattern is visible across many political cycles, though not mechanically true in every administration.

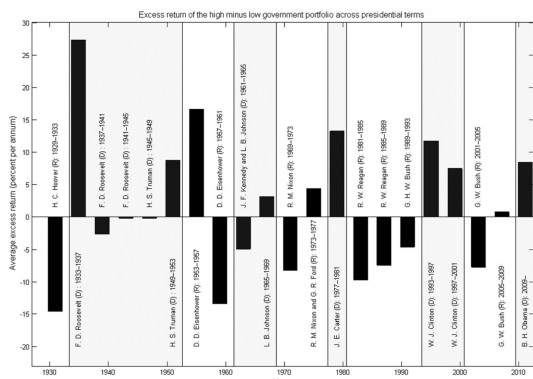


Fig. 1. Government exposure spread across presidential terms. The figure plots the average annualized value-weighted excess returns of the government exposure spread portfolio across presidential terms. The sample data are monthly from July 1929 to December 2009.

Table 7
Average returns of double-sorted portfolios on government exposure and geographical location.

This table reports the time series average of the excess value-weighted annualized returns of double-sorted portfolios on firms' government exposure and geographical location (US state). A firm belongs to a low (high) federal spending state if it is located in a state with per capita federal spending below (above) the median per capita federal spending across all US states. The results are reported across Democratic (Dem) and Republican (Rep) presidential terms. "Dif Dem-Rep" is the difference in the portfolio average excess returns across Democratic and Republican presidential terms, and "Dif p-value" is the corresponding Newey and West corrected p-value (in parenthesis). "Hi-Lo" is the return of the high minus low government exposure spread portfolio. The sample is from July 1953 to December 2009.

Portfolio	Low federal spending states				High federal spending states			
	Dem years	Rep years	Dif Dem-Rep	Dif p-value	Dem years	Rep years	Dif Dem-Rep	Dif p-value
Lo	8.16	5.98	2.18	(0.63)	7.83	7.70	0.13	(0.98)
2	7.49	4.04	3.45	(0.42)	8.06	5.80	2.26	(0.65)
3	9.36	3.78	5.58	(0.14)	11.50	2.72	8.78	(0.06)
4	9.13	2.52	6.60	(0.14)	11.81	1.00	10.81	(0.02)
Hi	14.08	4.38	9.71	(0.09)	14.78	0.04	14.73	(0.01)
Hi-Lo	5.92	-1.60	7.52	(0.09)	6.95	-7.66	14.61	(0.00)
p-value	(0.12)	(0.48)			(0.03)	(0.00)		

5.8 Table 7: Double-sorted portfolios on exposure and geography

Read.

- In low-federal-spending states, the high-minus-low government exposure spread is **+5.92%** under Democrats and **-1.60%** under Republicans.
- In high-federal-spending states, the spread is **+6.95%** under Democrats and **-7.66%** under Republicans.
- The Democratic-minus-Republican difference is **7.52%** in low-federal-spending states and **14.61%** in high-federal-spending states.

Economic message. The cross-sectional return reversal is strongest where federal spending is most economically relevant. This reinforces the government-spending interpretation.

5.9 Table 8: Cross-sectional return regressions

Read.

- The coefficient on $Gov \times Dem$ is positive and statistically significant across specifications.
- The coefficient on $Gov \times Rep$ is negative and statistically significant across specifications.
- With firm controls, the Democratic coefficient is about **1.00** and the Republican coefficient is about **-0.58**.
- Adding political contributions barely changes the pattern: the Democratic coefficient is **0.94** and the Republican coefficient is **-0.63**.
- Interactions with state federal-spending dependence show that both the positive Democratic effect and the negative Republican effect are stronger in states receiving more federal funds.

Economic message. Government exposure has predictive power beyond standard return characteristics and beyond a simple political-connectedness proxy.

Table 8

Cross-sectional regressions.

This table reports the results of Fama and MacBeth cross-sectional regressions of monthly excess stock returns on lagged firm characteristics. "Gov" is the firm-level exposure to the government sector, "Size" is the natural log of firm size, "BM" is the natural log of firm book-to-market ratio, "Mom" is firm momentum (prior cumulated 12 months returns), and "Beta" is the firm-level current market beta computed from daily returns within the month. "Dem" and "Rep" are dummy variables equal to one if the president is from the Democratic or Republican Party, respectively. "Political contrib" is the ratio of firms' political contribution to total assets. "Fed" denotes the per capita federal spending in a firm's headquarter state. It is measured as the cross-sectional percentile rank across all US states in each year. The table reports the average loadings for each cross-sectional regression and the corresponding Newey and West corrected p -value (in parenthesis). The sample data are monthly from July 1955 to December 2009.

Row	Intercept	Size	BM	Mom	Beta	Dem $\times$ Gov	Rep $\times$ Gov	Political contrib	Fed $\times$ Dem $\times$ Gov	Fed $\times$ Rep $\times$ Gov
1	1.02 (0.00)					1.53 (0.01)	-0.71 (0.06)			
2	0.99 (0.00)	-0.07 (0.06)	0.13 (0.00)	1.00 (0.00)	0.34 (0.00)	1.00 (0.02)	-0.58 (0.05)			
3	1.01 (0.00)	-0.07 (0.04)	0.13 (0.00)	1.00 (0.00)	0.34 (0.00)	0.94 (0.03)	-0.63 (0.04)	0.38 (0.00)		
4	1.02 (0.00)					1.26 (0.06)	-0.23 (0.64)		1.09 (0.10)	-0.91 (0.03)
5	1.20 (0.00)	-0.09 (0.01)	0.18 (0.00)	1.08 (0.00)	0.29 (0.00)	0.90 (0.10)	-0.22 (0.57)		0.53 (0.37)	-0.80 (0.04)

Table 9

Stock return predictability regressions of the government exposure portfolios.

This table reports monthly one-period-ahead predictability regressions of the annualized excess returns of the government exposure portfolios as well as of the government exposure spread portfolio (Hi-Lo) on a set of variables including two presidential dummy variables (Dem and Rep) equal to one if at the beginning of the month the president is from the Democratic or Republican Party, respectively, default spread (DEF), term spread (TERM), short-term Treasury bill rate (TB), dividend yield (DIV), inflation rate (Inflation), unemployment rate (Unemployment), and growth rate in industrial production (IP). All control variables are standardized with mean zero and unit standard deviation. We also report the difference in the slope coefficient of the Democratic and Republican dummies (Dif). p -values with Newey and West correction are reported in parenthesis. The sample data are monthly from July 1955 to December 2009.

Variable	Government exposure portfolios					
	Low	2	3	4	High	Hi-Lo
Political						
Dem	12.26 (0.00)	10.52 (0.00)	11.59 (0.00)	13.14 (0.00)	18.67 (0.00)	6.41 (0.05)
Rep	3.84 (0.17)	2.16 (0.44)	2.39 (0.36)	-0.32 (0.91)	-1.05 (0.76)	-4.89 (0.02)
Dif	8.43 (0.07)	8.37 (0.08)	9.20 (0.04)	13.46 (0.00)	19.72 (0.00)	11.29 (0.00)
Financial business cycle						
Def	5.26 (0.22)	4.60 (0.29)	2.06 (0.60)	7.55 (0.07)	7.55 (0.17)	2.29 (0.41)
Term	2.88 (0.45)	5.22 (0.15)	2.17 (0.53)	3.51 (0.35)	5.52 (0.20)	2.64 (0.32)
RF	-2.66 (0.42)	-3.90 (0.24)	-4.13 (0.21)	-4.65 (0.15)	-4.66 (0.28)	-1.99 (0.46)
Div	6.54 (0.03)	7.40 (0.01)	7.50 (0.01)	7.16 (0.02)	9.70 (0.03)	3.16 (0.35)
Macroeconomic business cycle						
Inflation	-3.92 (0.14)	-3.15 (0.21)	-3.19 (0.18)	-2.72 (0.30)	-4.85 (0.12)	-0.93 (0.62)
Unemployment	0.90 (0.83)	-1.25 (0.76)	1.71 (0.67)	-2.66 (0.52)	-2.14 (0.67)	-3.04 (0.30)
IP	1.83 (0.65)	4.43 (0.31)	5.74 (0.08)	6.36 (0.09)	7.59 (0.11)	5.76 (0.01)

5.10 Table 9: Business-cycle adjusted return predictability

Read.

- The high-minus-low spread remains **+6.41%** under Democratic presidencies after controlling for business-cycle variables.
- It remains **-4.89%** under Republican presidencies.
- The difference is **11.29%** and highly significant.
- The Democratic-minus-Republican effect is much larger for the high-exposure portfolio than the low-exposure portfolio: **19.72%** versus **8.43%**.

Economic message. The presidential-cycle return pattern is not explained by dividend yield, default spread, term spread, risk-free rate, inflation, unemployment, or industrial production growth.

5.11 Table 10: Presidential investment premium and abnormal returns

Read.

- The political-cycle strategy earns an average return of **5.31%** per year and a Sharpe ratio of **0.40**.
- Unconditional factor-model alphas are positive across CAPM, Fama–French, and Carhart models.
- When business-cycle instruments allow factor loadings to vary, abnormal returns are as large as **6.91%** per year.
- Adding a Democratic dummy to the conditioning set reduces alphas but does not eliminate them; alphas remain above **3.9%** in all models for the full sample.
- By year of the presidency, the average return is **8.28%** in year 2 and **7.17%** in year 3, but only **4.69%** in year 1 and **1.35%** in year 4.

Economic message. Standard risk factors do not fully explain the strategy's returns. The within-term timing is important: returns build up after policies begin to affect fundamentals, rather than appearing only immediately after election outcomes.

5.12 Table 11: Government exposure and cash-flow mean and volatility

Read.

- Future ROE is positively related to government exposure under Democrats and negatively related under Republicans.

Table 10

Presidential investment premium: realized and abnormal returns.

This table reports the average realized value-weighted returns (Ret), the Sharpe ratio, and abnormal returns (α) of the political cycle investment strategy, across different years of the political cycle: all years, first (Year 1), second (Year 2), third (Year 3), and fourth (Year 4) year of the presidential term. The abnormal returns of the political cycle investment strategy are obtained from three alternative asset pricing models: the CAPM (capital asset pricing model, which includes only the market factor), the Fama and French (1993) three-factor model (the Market, SMB (small minus big), and HML (high minus low) factors), and the Carhart (1997) four-factor model (the Market, SMB, HML, and Momentum factors). For each asset pricing model, we report three sets of abnormal return measures, which vary in the set of instruments used to capture time variation in risk. α^1 uses no instruments, and it corresponds to the unconditional version of each asset pricing model; α^2 corresponds to the specification in which DP (dividend-price ratio), DEF (default spread), TBSM (term spread), and RF (risk-free rate) are used as instruments; and α^3 corresponds to the same specification in α^2 plus a Democratic dummy variable equal to one if the president is from the Democratic Party. p -values with Newey and West correction are reported in parenthesis. The sample data are monthly from July 1955 to December 2009.

Row	Year	Ret	Sharpe ratio	Abnormal returns										
				CAPM			Fama and French			Carhart				
				α^1	α^2	α^3	α^1	α^2	α^3	α^1	α^2	α^3		
Entire sample period														
	1 All years	5.31 (0.00)	0.40	5.23 (0.01)	6.74 (0.00)	4.83 (0.00)	5.61 (0.00)	6.91 (0.00)	4.81 (0.00)	3.25 (0.07)	5.06 (0.01)	3.95 (0.00)		
Within the presidential cycle														
	2 Year 1	4.69 (0.20)	0.34	4.05 (0.21)	7.36 (0.06)	3.18 (0.36)	5.30 (0.16)	7.87 (0.03)	4.04 (0.22)	3.19 (0.41)	4.39 (0.21)	2.16 (0.50)		
	3 Year 2	8.28 (0.01)	0.72	8.30 (0.01)	9.59 (0.00)	9.06 (0.00)	9.39 (0.00)	9.96 (0.00)	8.32 (0.01)	6.98 (0.03)	8.01 (0.01)	7.25 (0.03)		
	4 Year 3	7.17 (0.02)	0.62	6.95 (0.03)	7.84 (0.01)	6.69 (0.03)	5.79 (0.01)	7.01 (0.03)	6.28 (0.06)	3.83 (0.22)	6.46 (0.03)	6.40 (0.03)		
	5 Year 4	1.35 (0.74)	0.08	1.30 (0.76)	2.41 (0.52)	0.65 (0.52)	1.89 (0.61)	2.82 (0.41)	0.46 (0.65)	-0.96 (0.79)	1.11 (0.73)	-0.21 (0.94)		

Table 11

Government exposure and cash flow mean and volatility across presidential terms.

This table examines the relation between cash flows (mean and volatility) and government exposure across presidential terms. The table reports the time series average slopes, with corresponding Newey and West corrected p -value in parenthesis, of several alternative specifications of the firm-level Fama and Macbeth cross-sectional regression

$$ROE_{it+1} = \alpha_0 + \alpha_1 \frac{V_{it}}{A_{it}} + \alpha_2 DD_{it} + \alpha_3 \frac{D_{it}}{B_{it+1}} + \alpha_4 ROE_{it} + \alpha_5 Gov_{it} \times Dem_{it} + \alpha_6 Gov_{it} \times Rep_{it} + \epsilon_{it+1},$$

where "ROE" is return-on-equity defined as net income to book-equity ratio (Row 1 to Row 3). "VA" is the ratio of market-value of assets to book-value assets. "DD" is a dummy variable for nondividend-paying firms. "DB" is the ratio of dividend payments to book-equity. "Gov" is the level of exposure to the government sector, and "Dem" and "Rep" are presidential dummy variables equal to one when the president is from the Democratic Party or Republican Party, respectively. Row 4 reports the results from a cross-sectional regression of ϵ_{it+1}^2 , where ϵ_{it} is the residual from the first stage cross-sectional regression on Row 3 on the presidential dummies interacted with the government exposure variable. "Diff Dem-Rep" is the difference in the government exposure slope coefficient between Democratic and Republican presidencies. The sample data are annual from 1955 to 2005.

Row	Intercept	VA	DD	DB	ROE	Gov × Dem	Gov × Rep	Diff Dem-Rep
Dependent variable: ROE_{it+1}								
1	0.12 (0.00)					0.04 (0.04)	-0.08 (0.00)	0.12 (0.00)
2	0.06 (0.00)	0.02 (0.02)	0.00 (1.00)	0.95 (0.00)		0.09 (0.00)	-0.06 (0.01)	0.15 (0.00)
3	0.03 (0.00)	0.01 (0.03)	0.00 (0.83)	0.37 (0.00)	0.50 (0.00)	0.04 (0.01)	-0.04 (0.00)	0.08 (0.00)
Dependent variable: ϵ_{it+1}^2								
4	0.05 (0.04)					0.07 (0.05)	0.05 (0.00)	0.02 (0.47)

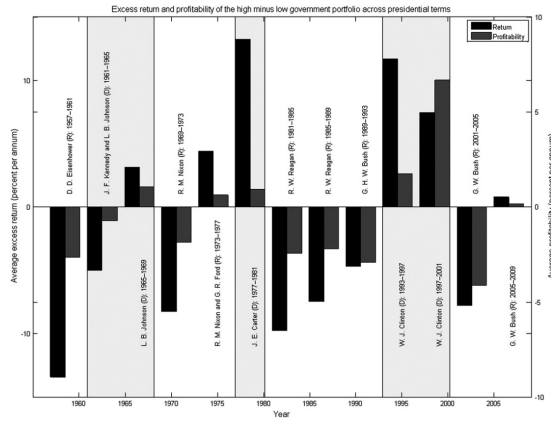


Fig. 2. Average returns and profitability of the government exposure spread across presidential terms. The figure plots the average annual profitability and value-weighted excess returns of the government exposure spread portfolio across presidential terms. The measure of profitability (return on equity) is adjusted to account for the control variables other than government exposure according to the regression specification in (6). The sample data are from July 1955 to December 2009.

- In the specification with lagged ROE, the $Gov \times Dem$ coefficient is **0.04**; the $Gov \times Rep$ coefficient is **-0.04**; their difference is **0.08** and statistically significant.
- For profitability volatility, the coefficient on $Gov \times Dem$ is **0.07** and the coefficient on $Gov \times Rep$ is **0.05**.
- The volatility difference across parties is only **0.02** and is not statistically significant.

Economic message. The cash-flow mean results line up strongly with the return results. The volatility results provide only weak support for a risk-based story, because volatility is higher for exposed firms under both parties.

5.13 Figure 2: Returns and profitability of the spread portfolio

Read.

- The figure plots average returns and adjusted profitability of the high-minus-low government exposure portfolio across presidential terms.
- The profitability pattern closely follows the return pattern.
- High-exposure firms tend to be more profitable than low-exposure firms under Democrats and less profitable under Republicans.

Economic message. This figure connects the asset-pricing result to real operating performance. The return pattern is not just a price pattern; it is also visible in future fundamentals.

6 Short summary

- The paper studies whether government spending exposure helps explain predictable stock returns across presidential partisan cycles.
- It constructs an industry-level exposure measure using input-output tables and the Leontief inverse.
- High-exposure firms outperform low-exposure firms under Democratic presidents but underperform under Republican presidents.
- The effect is stronger in states with high per capita federal spending.
- Firm characteristics, political contributions, business-cycle controls, and standard factor models do not explain the pattern.
- A political-cycle investment strategy earns about **5.3%** per year, with conditional alphas as large as **6.9%**.
- Future profitability moves in the same direction as returns, supporting an expected-cash-flow channel.
- Profitability volatility does not fully match the risk explanation, especially under Republicans.
- The authors interpret the evidence as market underreaction to predictable partisan variation in government-spending effects.

7 Conclusion

7.1 Core conclusion

Belo, Gala, and Li show that presidential partisan cycles matter for the cross section of stock returns through firms' exposure to government spending. The key result is a sharp sign reversal: government-exposed firms earn higher returns under Democrats and lower returns under Republicans.

7.2 Economic intuition

Government spending affects demand for some industries more than others. When Democratic administrations are associated with higher or more favorable government spending for exposed firms, these firms experience stronger future profitability and higher realized returns. When Republican administrations are associated with weaker spending support for exposed firms, the pattern reverses.

7.3 Asset-pricing implication

The result is difficult to reconcile with standard risk-factor models. The political-cycle strategy earns economically meaningful abnormal returns, and those returns appear gradually within presidential terms. This suggests that markets do not fully and immediately price the predictable consequences of partisan government-spending policies.

7.4 One-sentence takeaway

The paper turns political cycles into a cross-sectional asset-pricing test: firms more exposed to government spending behave like partisan fiscal-policy stocks, earning high relative returns under Democrats and low relative returns under Republicans because their expected profitability changes with government spending policy.